EXPRIMENT-8

8.)Write a Python program to implement a monoalphabetic substitution cipher using the keyword CIPHER to generate the cipher alphabet.

PROGRAM:

```
import string

keyword = "CIPHER"
plain = string.ascii_uppercase
cipher = ""

for ch in keyword + plain:
    if ch not in cipher:
        cipher += ch

message = input("Enter plaintext: ").upper()
encrypted = ""

for ch in message:
    if ch in plain:
        encrypted += cipher[plain.index(ch)]
    else:
        encrypted += ch

print("Encrypted text:", encrypted)

decrypted = ""
for ch in encrypted:
    if ch in cipher:
        decrypted += plain[cipher.index(ch)]
```

OUTPUT:

```
Enter plaintext: vishalli
Encrypted text: VDSBCJJD
Decrypted text: VISHALLI
```